SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO3_AT1 — EXPERIMENT

Course Code:	DSA06	**Course Title:**	Data Handling and Visualization
CO Assessed:	CO3 – Construction (BL3)	**Assessment:**	Assessment Tool 1 – Experiment
Weightage:	20% of CIA	**Total Marks:**	25 (5 Questions × 5 Marks)
CO3 Statement:	*Apply appropriate experimental charting techniques — pie charts, bean plots, violin plots, box plots, and stacked bar charts — to construct accurate visualizations from given data.(BL3)*		
Student Name:	M.SAI HARSHHITHA	**Reg. No.:**	192424408
Section / Batch:	B. TECH AI & DS	**Date:**	11.08.2026

Code of Conduct

I M.SAI HARSHITHA/192424408 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: M.SAI HARSHITHA

Instructions

- This test contains 5 questions, each carrying 5 marks. Total: 25 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 75 minutes.
- Graph paper or a ruler is recommended for accurate, to-scale drawing.
- Show all calculations (e.g., pie chart angles, five-number summary, IQR) as part of your answer.
- Every chart must include a title, correctly labeled axes (where applicable), and a legend where relevant.

Annexure A – EXPERIMENT

Rubrics for Calculation **ASSESSMENT RUBRIC**

AT1 — Experiment

CO3 · BT3: Apply ·

This rubric is used to assess student responses to the CO3-AT1 Experiment, in which students are given a small dataset and must construct an accurate pie chart, bean plot, violin plot, box plot, or stacked bar chart.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	**25%**	Demonstrates thorough, accurate understanding of the correct chart type (pie, bean, violin, box, or stacked bar) and the elements required to construct it for the given data.	Good understanding of the required chart type and construction elements, with only minor gaps.	Basic understanding of chart construction, but with some misconceptions about the correct chart type or required elements.	Poor understanding of core construction concepts; selects a chart type fundamentally unsuited to the data.
Explanation	**25%**	Shows detailed, clearly worked calculations (e.g., pie chart angles, five-number summary, IQR, stacked totals) and explains each construction step.	Shows clear calculations and construction steps, with adequate explanation for most parts.	Calculations or construction steps are shown but incomplete, or explanations are superficial.	Little or no working shown; calculations and construction steps are missing or unexplained.
Application	**20%**	Effectively applies chart construction techniques to produce a complete, accurate chart correctly matched to the data.	Applies construction techniques to produce a correct chart, with only minor errors.	Limited application of construction techniques; the chart partially represents the data but has notable errors.	Fails to apply appropriate construction techniques; the chart does not correctly represent the data.
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning

Organization	**15%**	The chart is neatly and logically constructed, with a title, correctly labeled axes or slices, a legend where applicable, and elements in a clear order.	The chart is well organized, with only minor issues in labeling, legend, or layout.	Basic construction; required elements (title, labels, legend) are present but poorly arranged or partly missing.	Poorly constructed; the chart is missing key elements such as a title, labels, or a legend.
Accuracy	**10%**	The chart is completely accurate — all values, proportions, quartiles, and calculations correctly match the given data.	Mostly accurate, with only minor omissions or a small error in one value or calculation.	Partially correct; some plotted values, proportions, or calculations are missing or incorrect.	Largely inaccurate; major errors in plotted values, proportions, or calculations relative to the given data.

Scoring Guide

For each criterion, award a performance level (1–4) based on the descriptors above. Multiply the level achieved (as a percentage of 4, i.e., Level 4 = 100%, Level 3 = 75%, Level 2 = 50%, Level 1 = 25%) by the criterion's weightage, then sum across all five criteria to obtain the total score out of 100%.

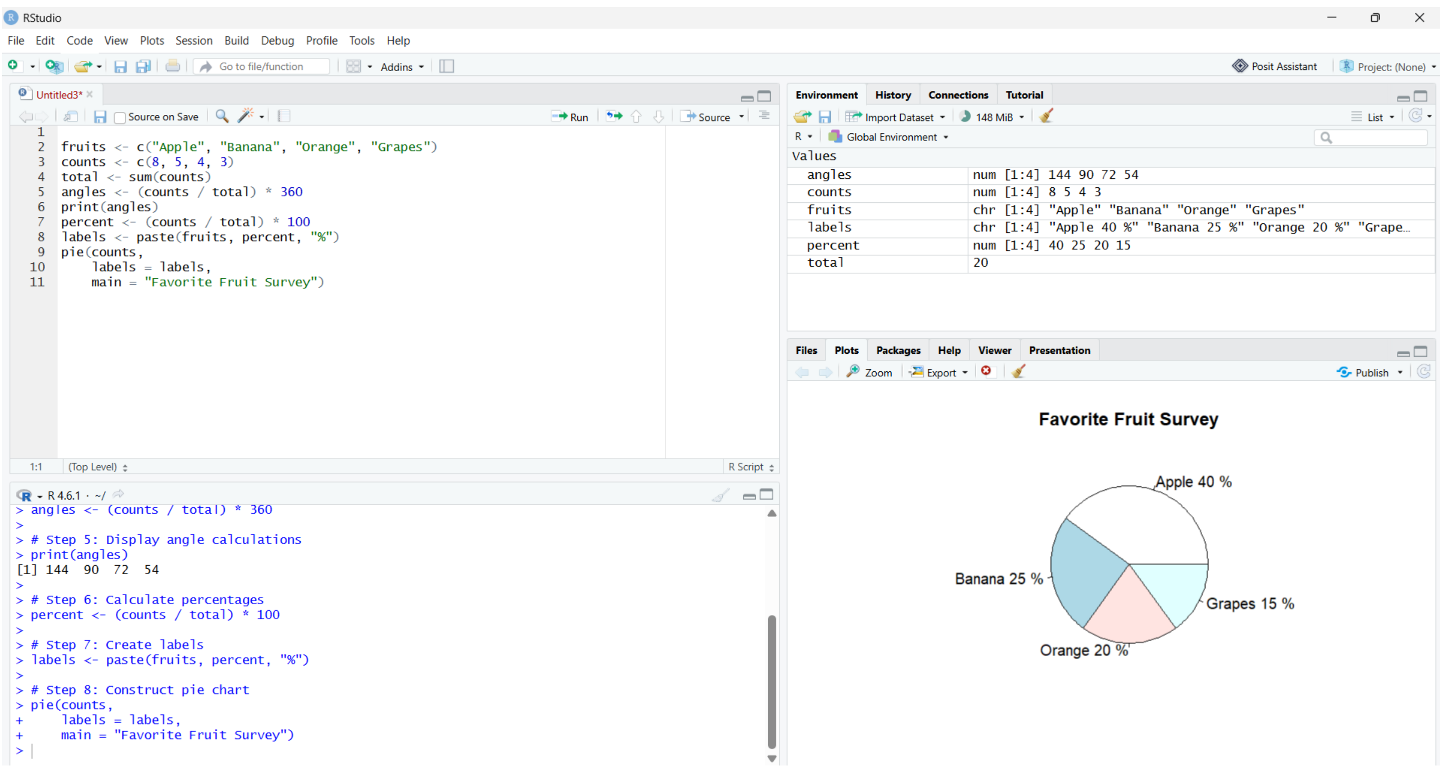
Faculty In-charge	Course Coordinator	Head of Department
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________

Pie Chart

Problem 1 (5 Marks)

Data: Favorite fruit survey of 20 people: Apple = 8, Banana = 5, Orange = 4, Grapes = 3.

Task: Construct a pie chart representing this data. Show your calculation of the angle (in degrees) for each fruit.



Interpretation of the Pie Chart

- The pie chart represents the favorite fruit preferences of **20 people**. The data consists of four fruits: Apple, Banana, Orange, and Grapes.
- **Apple** is the most preferred fruit, with **8 people**, representing **40%** of the total respondents. Its corresponding sector angle is **144°**.
- **Banana** is preferred by **5 people**, representing **25%** of the respondents. The corresponding sector angle is **90°**.
- **Orange** is preferred by **4 people**, representing **20%** of the respondents. Its corresponding sector angle is **72°**.
- **Grapes** is preferred by **3 people**, representing **15%** of the respondents. Its corresponding sector angle is **54°**.
- The pie chart clearly shows that **Apple has the largest share**, while **Grapes has the smallest share** among the four fruits.

Result

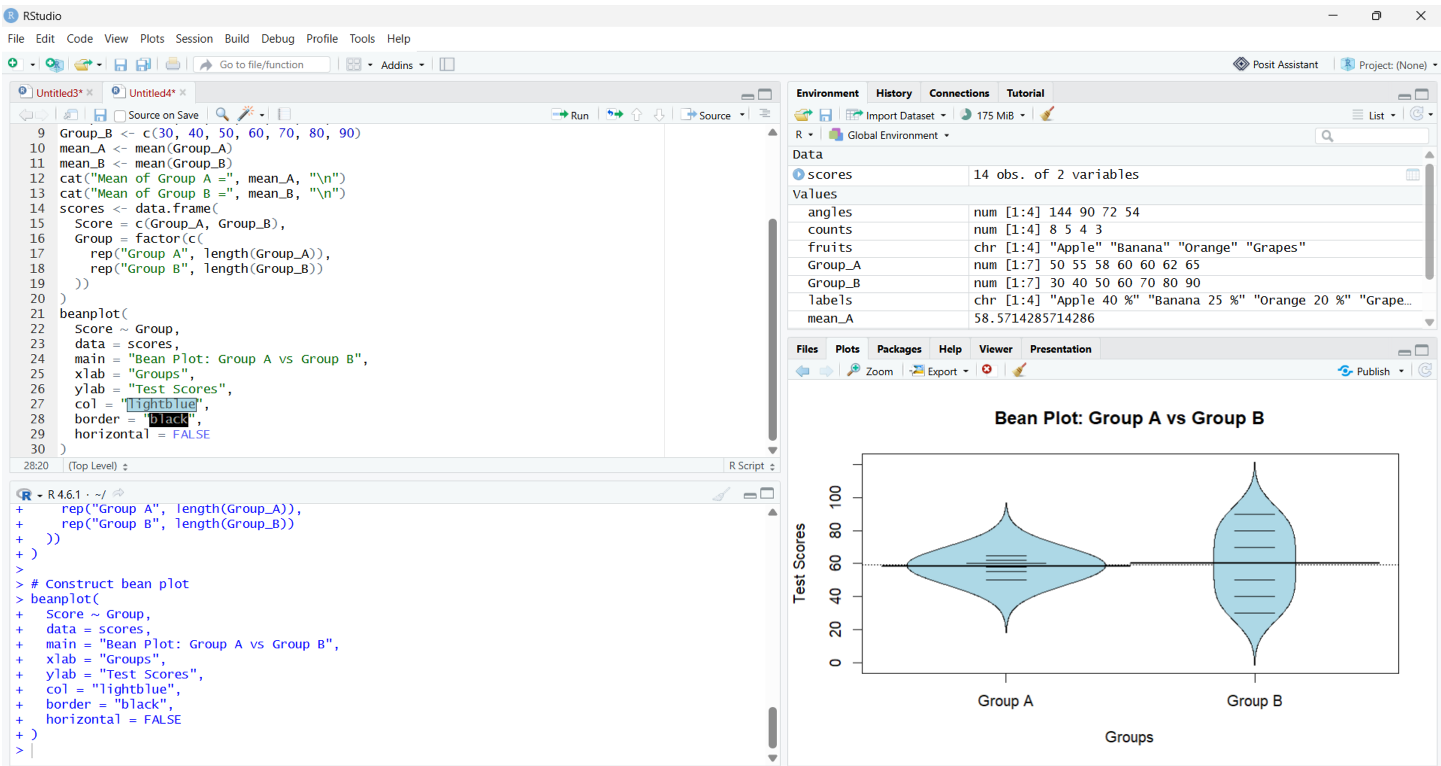
Thus, the pie chart was successfully constructed in R for the favorite fruit survey of 20 people. Apple has the highest preference with **40%**, followed by Banana with **25%**, Orange with **20%**, and Grapes with **15%**. The calculated sector angles are **144°, 90°, 72°, and 54°**, respectively, and their total is **360°**.

Bean plot

Problem 7 (5 Marks)

Data: Two groups' test scores — Group A: 50, 55, 58, 60, 60, 62, 65. Group B: 30, 40, 50, 60, 70, 80, 90.

Task: Construct bean plots for both groups and compare how their spread differs, even though both have a similar average.



Interpretation of the Bean Plot

- The bean plot represents the test score distributions of **Group A and Group B**, with 7 students in each group.
- **Group A** has scores ranging from **50 to 65**, with a mean score of **60** and a median score of **60**. The first quartile (Q1) is **56.5**, while the third quartile (Q3) is **61**, indicating that most of the scores are concentrated closely around 60.
- **Group B** has scores ranging from **30 to 90**, with a mean score of **60** and a median score of **60**. The first quartile (Q1) is **40**, while the third quartile (Q3) is **80**, showing a much wider spread of scores.
- Although both groups have a **similar average of 60**, their variability is considerably different. Group A has a range of only **15 marks**, whereas Group B has a range of **60 marks**.
- The bean plot clearly shows that **Group A's scores are more concentrated**, while **Group B's scores are widely distributed** across the score range.

Result

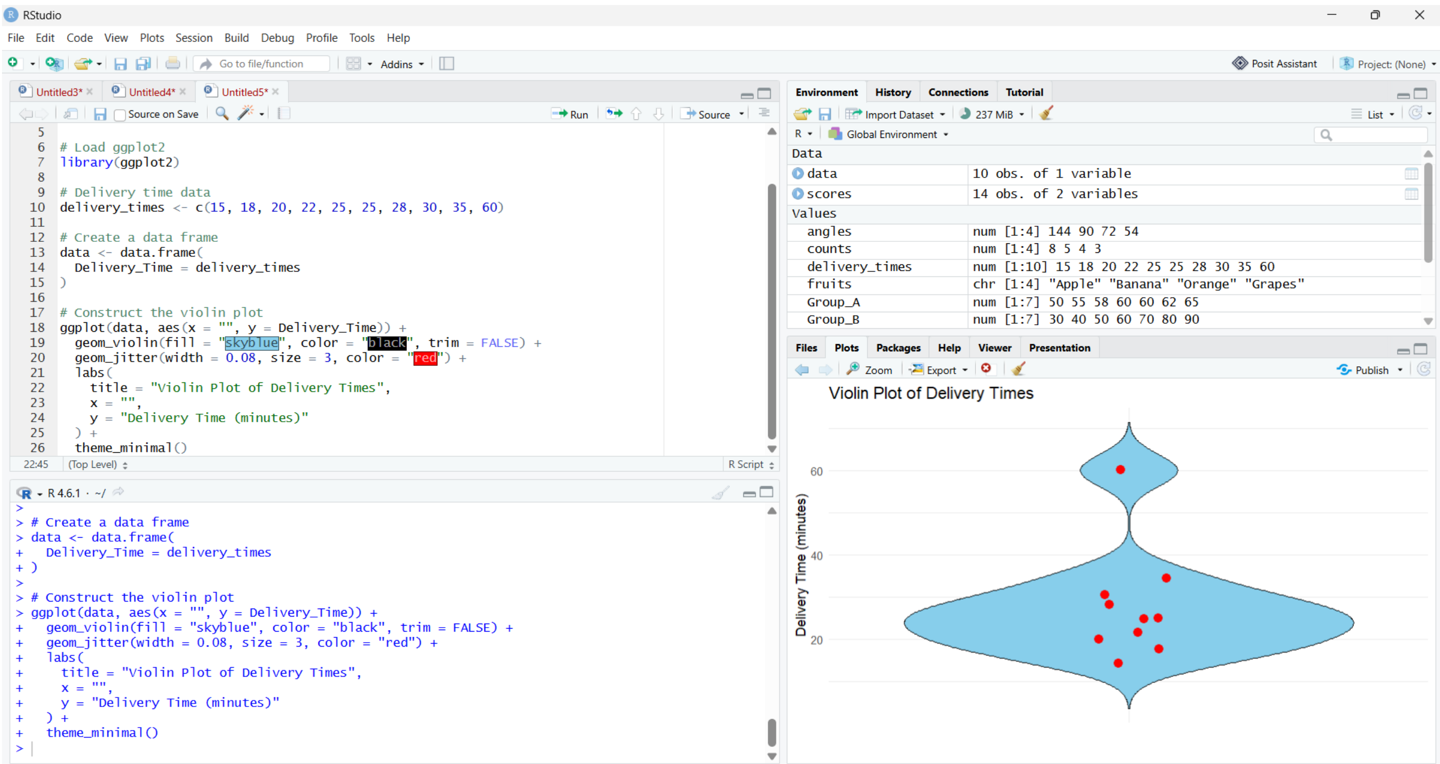
Thus, the bean plots were successfully constructed for both groups. **Both Group A and Group B have the same mean score of 60, but Group B has much greater spread and variability than Group A.** Group A shows consistent performance, whereas Group B shows considerable variation in test scores.

Violin plot

Problem 13 (5 Marks)

Data: Delivery times (in minutes) for 10 orders: 15, 18, 20, 22, 25, 25, 28, 30, 35, 60.

Task: Construct a violin plot and identify where most delivery times are concentrated.



Interpretation of the Violin Plot

- The violin plot represents the **delivery time distribution of 10 orders**.
- The first quartile (Q1) is **20.5 minutes**, indicating that approximately 25% of the delivery times are below this value.
- The median delivery time is **25 minutes**, meaning that half of the orders were delivered in less than 25 minutes and half took more than 25 minutes.
- The third quartile (Q3) is **29.5 minutes**, indicating that approximately 75% of the delivery times are below this value.
- Most delivery times are **concentrated between approximately 20 and 35 minutes**, with the widest part of the violin showing greater concentration around **25 minutes**.
- The **60-minute delivery time** is considerably higher than the main group and appears as a **potential outlier**.

Result

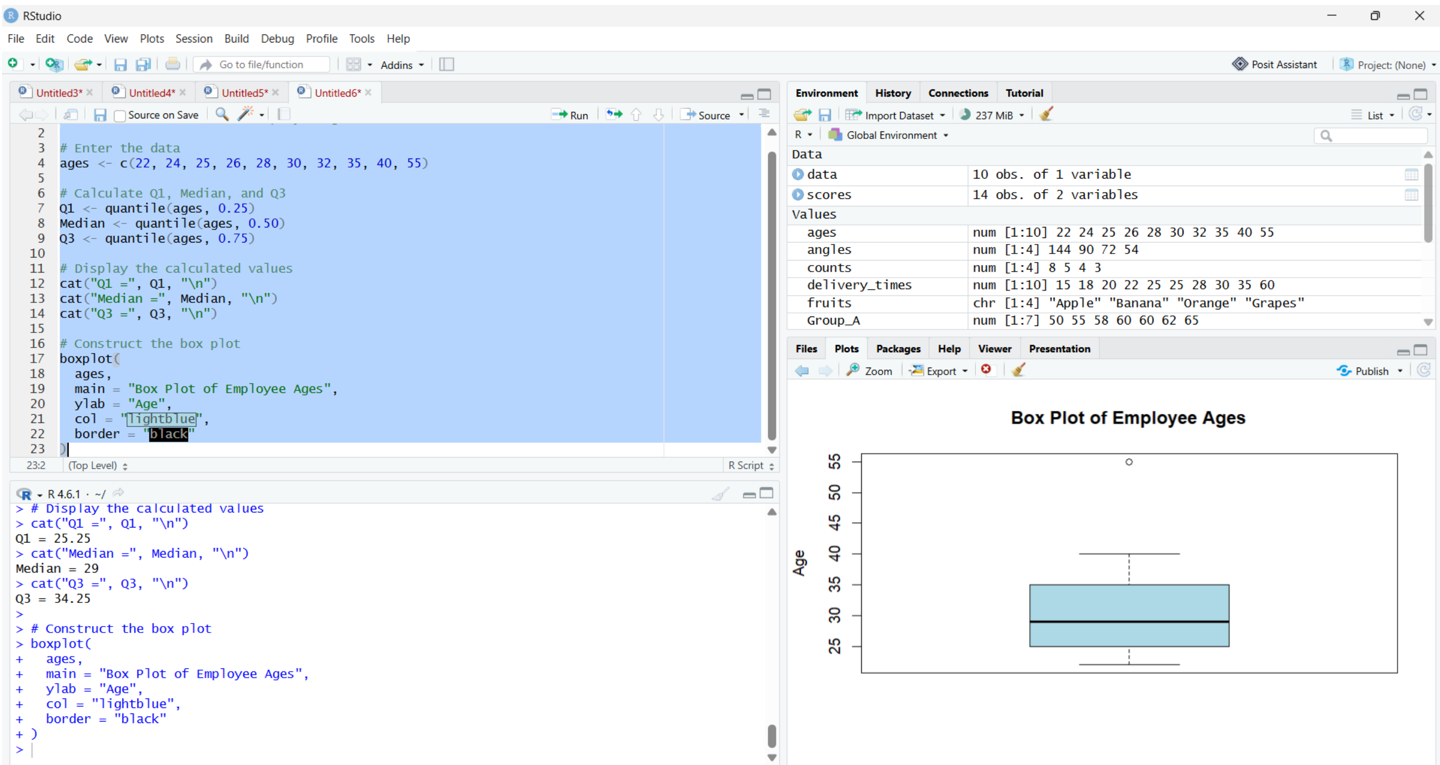
Thus, the violin plot was successfully constructed to represent the distribution of delivery times. The **Q1, median, and Q3** were calculated as **20.5 minutes, 25 minutes, and 29.5 minutes**, respectively. Most delivery times are concentrated between approximately **20 and 35 minutes**, while the **60-minute delivery** is an unusually high value.

Box plot

Problem 19 (5 Marks)

Data: Ages of 10 employees in a department: 22, 24, 25, 26, 28, 30, 32, 35, 40, 55.

Task: Construct a box plot for this data, calculating Q1, the median, and Q3.



Interpretation of the Box Plot

The box plot represents the age distribution of 10 employees in the department.

- The **first quartile (Q1)** is **25.25 years**, indicating that approximately 25% of the employees are younger than this age.
- The **median age** is **29 years**, meaning that half of the employees are younger than 29 and half are older.
- The **third quartile (Q3)** is **34.25 years**, indicating that approximately 75% of the employees are younger than this age.
- Most employee ages are concentrated between approximately **25 and 34 years**.
- The **55-year-old employee** is considerably older than the main group and appears as a potential outlier.

Result

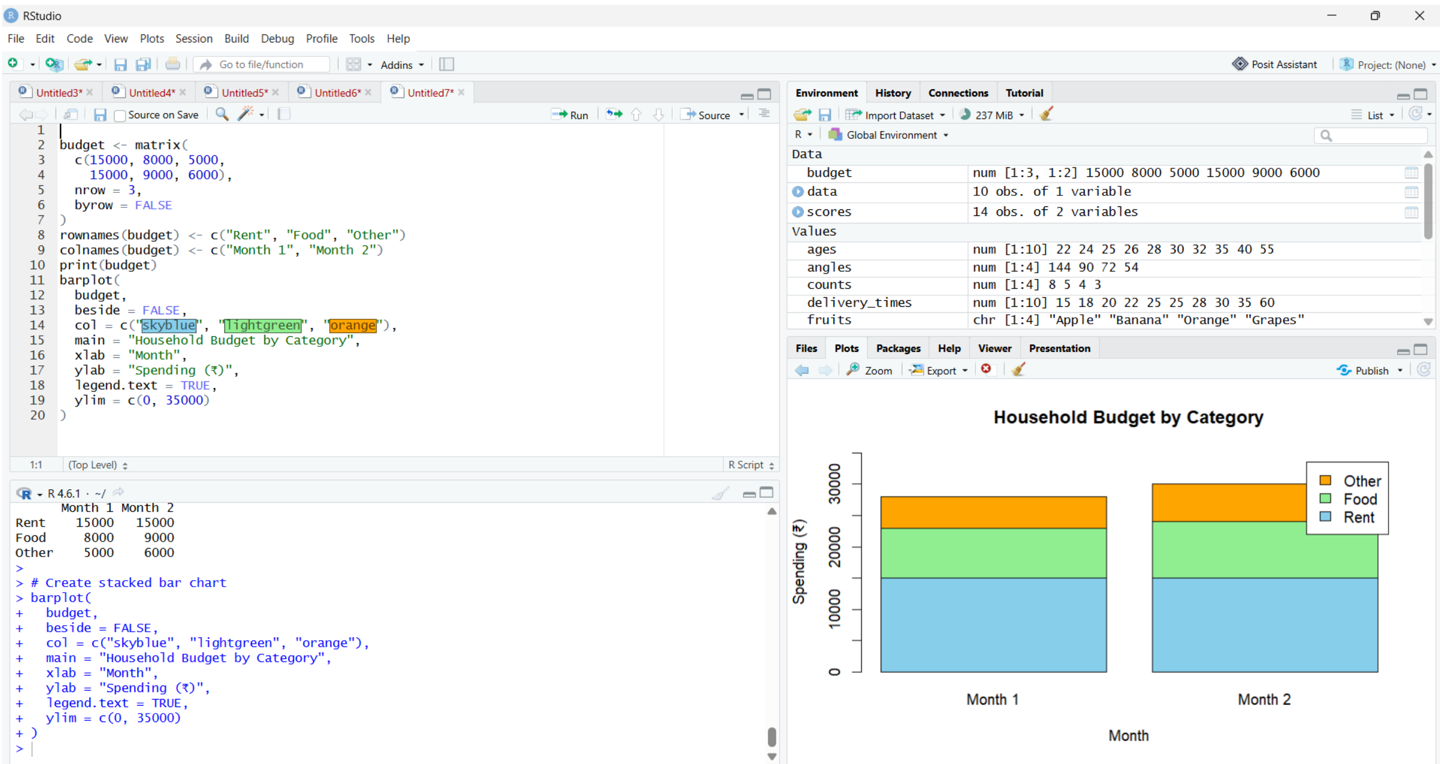
Thus, the box plot was successfully constructed to represent the distribution of employee ages, and Q1, median, and Q3 were calculated as 25.25 years, 29 years, and 34.25 years, respectively.

Stacked plot

Problem 25 (5 Marks)

Data: Household budget by category across 2 months: Month 1 — Rent = 15,000, Food = 8,000, Other = 5,000. Month 2 — Rent = 15,000, Food = 9,000, Other = 6,000.

Task: Construct a stacked bar chart showing total spending and category composition for both months.



Interpretation of the Stacked Bar Chart

The stacked bar chart represents the **household budget distribution across three categories—Rent, Food, and Other—for two months**.
In Month 1, the total household expenditure is **₹28,000**, consisting of ₹15,000 for Rent, ₹8,000 for Food, and ₹5,000 for Other expenses. In Month 2, the total expenditure increases to **₹30,000**, with ₹15,000 spent on Rent, ₹9,000 on Food, and ₹6,000 on Other expenses.

- Rent remains the **highest expenditure category** in both months.
- Food and Other expenses show an increase of **₹1,000 each** from Month 1 to Month 2.
- Overall, the chart clearly illustrates the **total monthly spending as well as the composition of expenditure by category**.

Result

Thus, the stacked bar chart was successfully constructed to compare the total household spending and category-wise expenditure for the two months.